

Report R15 — the tower in the attractor plane, and whether the resets are idle inside it

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

In the right-hand panel of R9's F1 a column of V+reset rules stands at $a_{\text{att}} = 1$ with b_{att} running from 1.41 to 1.89: a bounded number of attractors, the largest of which still grows exponentially. There are 68 such rules of the 160 V+reset ones. Their growth bases are not spread out — 61 of the 68 sit on five *exact* algebraic constants, φ and $\sqrt{3}$ taking 21 each. The natural explanation is that the dissipative channels never act inside the largest attractor, so what grows there is really the underlying unitary. **That is true for exactly 12 of the 68, and false for the other 56.** For the twelve it is true in the strongest sense: the largest attractor *is* the largest sector of the coherent parent, state for state, and all twelve sit at φ . For the other fifty-six the resets do work at $\Theta(N)$ sites — the count of firing sites grows linearly with system size, so this is not a boundary effect — and the exponential growth survives dissipation rather than avoiding it.

The mechanism common to both is a subshift. The largest attractor is a subshift of finite type over the two-site brick-wall unit cell: membership is decided by which pairs of adjacent 2-site blocks occur, and $b_{\text{att}} = \sqrt{\lambda_{\max}(T)}$ for the 4×4 block-transition matrix T . Verified exactly — the grammar learned from the attractor regenerates the attractor and nothing else — for 62 of the 68 at two sizes, plus one more with an extra block of memory. A reset is a *local* map, so the set it leaves invariant is locally constrained, so it is counted by a transfer matrix whatever the map does to individual states; unitarity is sufficient for an algebraic base, not necessary. The five block Perron roots are 2, $1 + \sqrt{2}$, φ^2 , 3, $2 + \sqrt{2}$ and $(3 + \sqrt{17})/2$, all roots of quadratics. This also names a constant the numerics had left open, $\sqrt{1 + \sqrt{2}} = 1.5537740$, and leaves exactly one unidentified: the 1.718005 of rules 169 and 225, now bounded above by $\sqrt{3}$. Nine rules reach φ *with* firing resets, so the constant does not determine which of the two mechanisms is at work. obc0, $N = 10$ for the census.

1 The tower

A rule is in the tower if it is V+reset, its attractor *count* is flat ($a_{\text{att}} = 1$ to within 0.02) and its largest attractor grows exponentially ($b_{\text{att}} > 1.02$). That is 68 rules.

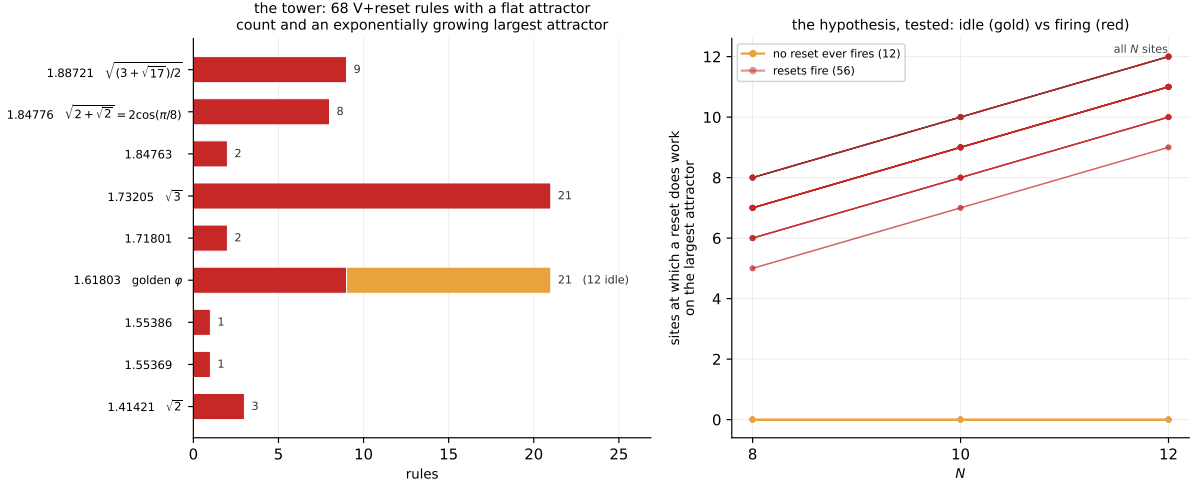


Figure 1: **Left:** the tower, one bar per distinct b_{att} , gold marking the rules on which no reset ever fires. **Right:** the hypothesis tested — the number of sites at which some reset does work on the largest attractor, against N . Flat at zero for twelve rules; linear in N for the rest, tracking the dotted “all N sites” line.

1.1 Where the bases sit

b_{att}	rules	exact	constant
1.414214	3	3	$\sqrt{2}$
1.553687	1	0	—
1.553861	1	0	—
1.618034	21	21	golden φ
1.718005	2	0	—
1.732051	21	20	$\sqrt{3}$
1.847632	2	0	—
1.847759	8	8	$\sqrt{2 + \sqrt{2}} = 2 \cos(\pi/8)$
1.887208	9	9	$\sqrt{(3 + \sqrt{17})/2}$

Five constants absorb 61 of the 68, and for those the base comes from an exact integer recurrence rather than a fit: $\sqrt{2}$ (3 rules), φ (21), $\sqrt{3}$ (20 of 21), $2 \cos(\pi/8) = \sqrt{2 + \sqrt{2}}$ (8) and $\sqrt{(3 + \sqrt{17})/2}$ (9). The remaining seven are fits. Two of them, at 1.847632, are within 1.3×10^{-4} of $2 \cos(\pi/8)$ and are almost certainly that constant imperfectly fitted; the pair at 1.5537/1.5539 and the pair at 1.7180 have no identification here and are left as fits rather than rounded into a name.

These are the constants of *constrained binary words* — Perron roots of small transfer matrices — which is exactly the company the unitary rules keep. That is what makes the idle-reset hypothesis so natural.

2 The hypothesis, tested

The dissipative channels never fire on the largest attractor, so what grows there is the underlying unitary.

A reset D writes 0 and E writes 1; either does *work* only when the target bit is not already at that value, and otherwise acts as the identity on that state. `tower.firing_sites` runs the brick-wall sweep by hand, following every Hadamard branch, and returns the sites at which

some branch had a reset do work. (The hand sweep is validated against the engine’s successor map in the tests: it must reproduce `wcc.make_succ` exactly, over all states of 37 rules at three sizes, or the firing flags mean nothing.)

rules		
no reset ever fires on the largest attractor	12	hypothesis holds
some reset fires	56	hypothesis fails

2.1 The twelve, and why they work

rule	tuple	b_{att}	$ A $	parent	parent D_{max}
1	VDDD	1.61803	144	W201	144
9	VIDD	1.61803	144	W201	144
65	VDID	1.61803	144	W201	144
110	IEIV	1.61803	55	W108	55
111	EEIV	1.61803	55	W108	55
124	IIEV	1.61803	55	W108	55
125	EIEV	1.61803	55	W108	55
126	IEEV	1.61803	55	W108	55
127	EEEV	1.61803	55	W108	55
129	VDDI	1.61803	144	W201	144
137	VIDI	1.61803	144	W201	144
193	VDII	1.61803	144	W201	144

For all twelve the statement holds in the strongest available form: the child’s successor map *coincides* with the coherent parent’s on every state of the largest attractor, and the attractor has exactly the size of the parent’s largest sector. It is not merely that the resets happen to be idle — the attractor *is* a sector of the parent, state for state.

They fall into two families of six, and the mechanism is visible in the tuple:

family	tuples	parent	$ A $ at $N = 10$
V at r_{00} , rest from {D,I}	VDDD, VIDD, VDID, VDDI, VIDI, VDII	W201	$144 = F_{12}$
V at r_{11} , rest from {E,I}	IEIV, EEIV, IIEV, EIEV, IEEV, EEEV	W108	$55 = F_{10}$

The reset’s target value agrees with what those sites already carry. In the first family the Hadamard fires only on the neighbourhood 00, and every D sits at a neighbourhood that occurs only when the site is already 0, so writing 0 changes nothing. The second family is the same statement with $0 \leftrightarrow 1$ and E for D. The surviving set is the parent’s Fibonacci-counted sector, hence φ — and indeed **every one of the twelve sits at φ** , none at any other constant.

2.2 The fifty-six: the resets fire throughout

For these the resets fire, and not marginally: the number of *sites* at which some reset does work on the largest attractor grows linearly with N (right panel of the figure), typically reaching $N - 1$ or N . So it cannot be rescued as a boundary effect or as a bounded set of defect sites — dissipation acts throughout the bulk of the attractor, and the exponential growth survives it rather than avoiding it.

2.3 So what *is* the mechanism? A subshift on the unit cell

The attractor is a **subshift of finite type over the two-site brick-wall unit cell**. Write a state as $N/2$ blocks of two sites. Then membership in the largest attractor is decided entirely

by *which pairs of adjacent blocks occur* — nothing longer-range is needed — and the growth base is

$$b_{\text{att}} = \sqrt{\lambda_{\text{max}}(T)},$$

with T the 4×4 zero-one matrix of allowed block-to-block transitions. The square root is because one block is two sites.

This is checked, not assumed. `tower.sft.check` learns the grammar *from* the attractor — the first blocks, the allowed adjacent pairs, the last blocks — regenerates every state the grammar permits, and asks whether that reproduces the attractor exactly. It does for **62 of the 68** at $N = 10$ and $N = 12$ simultaneously, and one more (rule 118) with a single extra block of memory. For all of them $\sqrt{\lambda_{\text{max}}}$ matches the reported base, with no exceptions.

	$\lambda_{\text{max}}(T)$	minimal polynomial	rules	$b_{\text{att}} = \sqrt{\lambda}$	constant
	2	$x - 2$	3	1.414214	$\sqrt{2}$
$1 + \sqrt{2} = 2.414214$		$x^2 - 2x - 1$	2	1.553774	$\sqrt{1 + \sqrt{2}}$
$\varphi^2 = 2.618034$		$x^2 - 3x + 1$	20	1.618034	φ
	3	$x - 3$	19	1.732051	$\sqrt{3}$
$2 + \sqrt{2} = 3.414214$		$x^2 - 4x + 2$	9	1.847759	$2 \cos(\pi/8)$
$(3 + \sqrt{17})/2 = 3.561553$		$x^2 - 3x - 2$	9	1.887208	—

The grammars, explicitly. Writing a block as its two site values and indexing T by 00, 10, 01, 11:

rule	b_{att}	allowed pairs	blocks used	the constraint
3	$\sqrt{3}$	9/16	00, 10, 01	the block 11 never occurs; the other three are free
11	φ	8/16	00, 10, 01	as above, and 00 cannot be followed by 10
191	$\sqrt{2}$	6/16	10, 01, 11	00 never occurs, and only two successors survive
167	$\sqrt{1 + \sqrt{2}}$	10/16	all four	
35	$2 \cos(\pi/8)$	14/16	all four	only $11 \rightarrow 10$ and $11 \rightarrow 11$ forbidden
19	$\sqrt{(3 + \sqrt{17})/2}$	14/16	all four	only $11 \rightarrow 00$ and $11 \rightarrow 01$ forbidden

Rule 3 is the cleanest and it covers the whole $\sqrt{3}$ group of 19: the surviving states are exactly those in which *the block 11 never appears*, with no further condition, so the attractor is all words over a three-letter block alphabet and $|A| = 3^{N/2}$ exactly — 729 at $N = 12$, as computed. At the other extreme rules 35 and 19 forbid only two of the sixteen block pairs each, and the difference between which two is the whole difference between $2 \cos(\pi/8)$ and $\sqrt{(3 + \sqrt{17})/2}$.

This is why the firing resets do not spoil anything. A reset is a *local* map, so the set it leaves invariant is cut out by local constraints, and a locally-constrained set of words is counted by a transfer matrix regardless of what the map does to individual states. Unitarity on the attractor is *sufficient* for an algebraic base — that is the twelve — but it is not necessary. Locality is what does the work, and every rule in the tower is local by construction.

Seen this way the two mechanisms are one mechanism at different strengths. The twelve inherit the parent’s transfer matrix outright because the resets write nothing; the fifty-six get a *different* transfer matrix, because the resets forbid block pairs the parent allowed. Both are subshifts on the same unit cell.

2.4 The two constants this identifies

The transfer matrix supplies exact values where the numerical fit gave only four digits. Of the seven tower bases that §1 lists as fits rather than exact recurrences, four are now pinned:

rules	fitted b_{att}	$\sqrt{\lambda_{\text{max}}}$	identification
167, 181	1.553861, 1.553687	1.5537740	$\sqrt{1 + \sqrt{2}}$ (new)
41	1.847632	1.8477590	$2 \cos(\pi/8)$
183	1.732051	1.7320508	$\sqrt{3}$

$\sqrt{1 + \sqrt{2}} = 1.5537740$ was one of the two unidentified pairs in §1; it is now named.

2.5 Five rules that are not subshifts at this order

Rules 97, 117, 123, 169 and 225 are not SFTs on blocks at order 1 or 2: the learned grammar regenerates *more* than the attractor. Since the grammar is learned from the attractor, the closure always contains it, so $\sqrt{\lambda_{\text{max}}}$ is an *upper bound* on the true base and the attractor is a proper subshift of it — sofic, or of longer range. Rules 169 and 225 are the sharpest case: closure base $\sqrt{3} = 1.732051$, measured 1.718005. That pair remains the one genuinely unidentified constant in the tower, now with a bound rather than nothing.

2.6 The constant does not identify the mechanism

Twenty-one rules sit at φ ; twelve are idle and **nine are not** — rules 11, 47, 81, 117, 139, 161, 171, 209 and 241 reach the same golden ratio with resets firing at $\Theta(N)$ sites. So one cannot read the mechanism off the growth base, which is worth saying because the temptation to do so is exactly what this report set out to test.

3 What this settles and what it does not

- **Settled.** The idle-reset explanation is a real mechanism and it is a minority one: 12 of 68, all at φ , all inheriting a parent sector outright.
- **Settled.** For the other 56 the resets act on $\Theta(N)$ sites of the attractor, so no version of “the channel is secretly unitary there” survives.
- **Settled** (§2.3). The transfer matrices, for 63 of the 68: the attractor is a subshift on the 2-site unit cell and $b_{\text{att}} = \sqrt{\lambda_{\text{max}}(T)}$, with the five Perron roots all quadratic irrationals. This is the mechanism for the firing rules and it subsumes the idle ones.
- **Settled.** One of the two unidentified pairs: 1.5537 is $\sqrt{1 + \sqrt{2}}$.
- **Open.** The 1.7180 pair (rules 169, 225), which is not a subshift at order 1 or 2 on blocks. Its SFT closure gives the bound $b_{\text{att}} < \sqrt{3}$, and the true value is 1.718005.
- **Open.** Rules 97, 117, 123 likewise fail the subshift test at this order; their attractors are sofic or longer-range.
- **Caveat.** obc0, and the census is at $N = 10$ with the firing-site scaling checked at $N = 8, 10, 12$. Attractors are monitored-only objects, so everything here is a statement about the measured chain.

Reproduce

```
python -m qca_fragmentation.viz.tower --bc obc0 --N 10
```

writes `fig_tower_obc0` and the two tables. The module is `code/qca_fragmentation/viz/tower.py`; `idle_census(rule, N)` returns the per-rule verdict and `firing_sites(x, N, t)` the raw site set for one state.